

## Binomial random variables



1 In a binomial experiment:

- a Are the trials independent or dependent?
- b How many possible outcomes are there for each trial?

2 State whether following scenarios describe a binomial random variable:

- a Rolling a six-sided die 20 times and counting the number of times the die lands on 1.
- b The sum of the outcomes of rolling a six-sided die 10 times.
- c Selecting 70 random people from the population and recording the number of females.
- d A particular coin is more likely to land on heads than tails. Tossing this coin 10 times and recording the number of times it lands on heads.
- e Drawing a marble with replacement from a bag containing purple, green and yellow marbles, and noting the number of purple marbles.
- f Drawing a marble with replacement from a bag containing purple, green and yellow marbles, and noting the number of red marbles.

3 Consider the binomial probability formula:

$$P(x) = {}^nC_x p^x (1-p)^{n-x}$$

where  $P(x)$  is the probability of obtaining  $x$  successes in  $n$  independent trials.

- a What does  $p$  represent in the formula?
- b What does  $(1-p)$  represent in the formula?

4 For a binomial probability distribution with  $n$ , trials and probability of success  $p$ , Ned performed the following calculation:

$$\binom{n}{2} p^2 \times 0.7^4$$

- a State the probability of success.
- b State the number of trials.
- c What probability is Ned trying to find with this calculation?

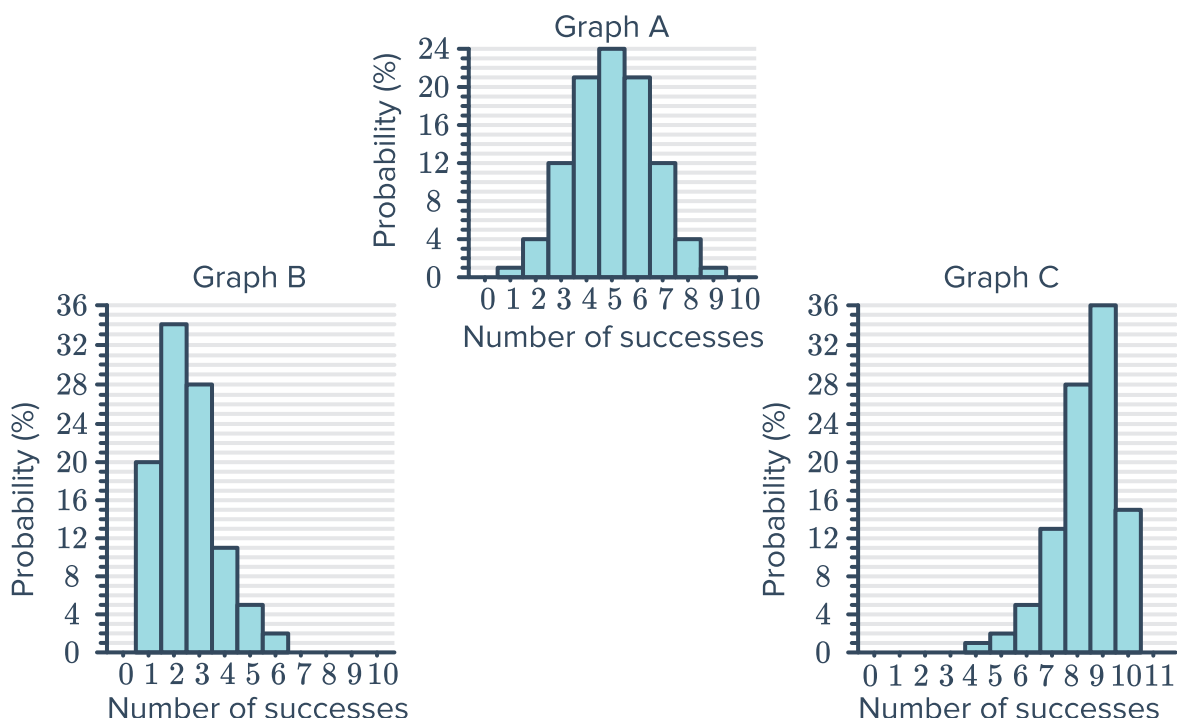
5 Consider the binomial power  $(p+q)^4$ .

- a Expand  $(p+q)^4$ .
- b We can interpret the terms in the expansion in terms of probabilities in a binomial distribution, with  $p$  the probability of success and  $q$  the probability of failure. What probability would the term  $\binom{4}{2} p^2 q^2$  in the expansion represent?



## Distributions of binomial random variables

- 6 Consider the three graphs below that show binomial distributions for  $n = 10$  and  $p = 0.15$ ,  $0.5$  and  $0.85$ :



- Which value of  $p$  corresponds to Graph A?
  - Which value of  $p$  corresponds to Graph B?
  - Which value of  $p$  corresponds to Graph C?
- 7 Consider a binomial distribution of a random variable  $X$  with  $p = 0.5$  and  $n = 17$ .
- Is  $P(X = 5)$  greater than, less than or equal to  $P(X = 12)$ ?
  - Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?
- 8 Consider a binomial distribution of a random variable  $X$  with  $p = 0.13$  and  $n = 11$ .
- Is  $P(X = 4)$  greater than, less than or equal to  $P(X = 7)$ ?
  - Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

9 Consider a binomial distribution of a random variable  $X$  with  $p = 0.83$  and  $n = 17$ .

- Is  $P(X = 2)$  greater than, less than or equal to  $P(X = 15)$ ?
- Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

10 For each of the following table of values for the binomial distribution  $X$ :

- Describe the shape of the graph as positively skewed, symmetric, or negatively skewed.
- Calculate the value of  $p$  correct to two decimal places.

a

| $x$        | 0     | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $P(X = x)$ | 0.004 | 0.031 | 0.109 | 0.219 | 0.274 | 0.219 | 0.109 | 0.031 | 0.004 |

b

| $x$        | 0     | 1     | 2   | 3    | 4     | 5     | 6     | 7 | 8 |
|------------|-------|-------|-----|------|-------|-------|-------|---|---|
| $P(X = x)$ | 0.152 | 0.323 | 0.3 | 0.16 | 0.053 | 0.011 | 0.001 | 0 | 0 |

c

| $x$        | 0 | 1 | 2     | 3     | 4     | 5     | 6     | 7     | 8     |
|------------|---|---|-------|-------|-------|-------|-------|-------|-------|
| $P(X = x)$ | 0 | 0 | 0.001 | 0.009 | 0.046 | 0.147 | 0.294 | 0.336 | 0.168 |

## Probability for binomial random variables

11 Find the value of the following:

- ${}^5C_4 \times (0.1)^4 \times 0.9 + {}^5C_5 \times (0.1)^5 \times (0.9)^0$
- ${}^5C_3 \times (0.8)^3 \times (0.2)^2 + {}^5C_4 \times (0.8)^4 \times 0.2 + {}^5C_5 \times (0.8)^5 \times (0.2)^0$

12 A binomial random variable  $X$  has 13 independent trials and a probability of success of 0.4.

- Using the definition for the binomial random variable  $X$ , write the formula for  $P(X = 2)$ .
- Calculate  $P(X = 2)$  to four decimal places.

13 For a binomial random variable  $X$ ,  $P(X = \text{[img alt="A small graphic of a basketball with a blue and orange pattern." data-bbox="475 21 525 58"]}) = {}^{10}C_3 \times 0.6^3 \times 0.4^7$ .

- a Calculate  $P(X = 3)$  to four decimal places.
- b State the number of trials in this experiment.
- c State the probability of success in this experiment.

14  $X$  is a binomial variable with the probability mass function:

$$P(X = k) = {}^4C_k \times (0.4)^k \times (0.6)^{4-k} \text{ for } k = 0, 1, 2, 3, 4$$

- a State the number of trials for this distribution.
- b State the probability of success.
- c Write an expression for  $P(X = 2)$ .
- d How many ways can we get 2 successes in the 4 trials?
- e Calculate the probability  $P(X = 2)$ .

15  $X$  is a binomial variable with the probability mass function:

$$P(X = k) = {}^3C_k \times (0.3)^k \times (0.7)^{3-k} \text{ for } k = 0, 1, 2, 3$$

Complete the table:

|            |       |       |   |   |
|------------|-------|-------|---|---|
| $x$        | 0     | 1     | 2 | 3 |
| $P(X = x)$ | 0.343 | 0.441 |   |   |

16 Consider the random variable  $X$ , which has distribution  $X \sim B(4, 0.1)$ .

- a Write an expression for  $P(X = 1)$ .
- b Write an expression for  $P(X = 2)$ .
- c Complete the following table:

|            |        |   |   |   |        |
|------------|--------|---|---|---|--------|
| $x$        | 0      | 1 | 2 | 3 | 4      |
| $P(X = x)$ | 0.6561 |   |   |   | 0.0001 |

17 A binomial random variable  $X$  is defined as:

$$P(X = x) = {}^{20}C_x \times 0.75^x \times 0.25^{20-x} \text{ for } x = 1, 2, \dots, 20$$

- a State the number of trials in this experiment.
- b State the probability of success in this experiment.
- c Calculate  $P(X = 11)$  to four decimal places.

$$P(X = k) = {}^6C_k \times (0.8)^k \times (0.2)^{6-k} \text{ for } k = 0, 1, 2, 3, 4, 5, 6$$

- a State the number of trials for this distribution.
- b What is the probability of failure?
- c Calculate the probability  $P(X = 3)$ .
- d Calculate the probability  $P(X \leq 3)$ .

19  $X$  is a binomial variable with the probability mass function:

$$P(X = k) = {}^5C_k \times (0.4)^k \times (0.6)^{5-k} \text{ for } k = 0, 1, 2, 3, 4, 5$$

- a Calculate  $P(X \geq 3)$ .
- b Calculate  $P(X < 3)$ .

20  $X$  is a binomial variable with the probability mass function:

$$P(X = k) = {}^5C_k \times (0.2)^k \times (0.8)^{5-k} \text{ for } k = 0, 1, 2, 3, 4, 5$$

- a State the number of trials for this distribution.
- b State the probability of success.
- c Calculate the probability  $P(X > 3)$ .
- d What is the most likely number of successes?
- e Calculate the mean of the distribution.
- f Calculate the variance of the distribution.

21 Consider the random variable  $X$ , which has distribution  $X \sim B\left(8, \frac{4}{5}\right)$ .

- a State the number of trials for this distribution.
- b State the probability of success.
- c Calculate the probability  $P(X > 4)$  to two decimal places.
- d What is the most likely number of successes?
- e Calculate the mean of the distribution.
- f Calculate the variance of the distribution.

**22**  $X$  is a binomial variable with a probability distribution given in the table:



| $x$        | 0       | 1       | 2      | 3      | 4       | 5       |
|------------|---------|---------|--------|--------|---------|---------|
| $P(X = x)$ | 0.00243 | 0.02835 | 0.1323 | 0.3087 | 0.36015 | 0.16807 |

**a** Find  $P(X \geq 4)$ .

**b** Find  $P(X < 3)$ .

**23** The following table represents a probability distribution of a binomial random variable:

| $x$        | 0      | 1      | 2      | 3      | 4      |
|------------|--------|--------|--------|--------|--------|
| $P(X = x)$ | 0.2401 | 0.4116 | 0.2646 | 0.0756 | 0.0081 |

**a** State the number of trials for this distribution.

**b** Calculate the probability  $P(X = 3)$ .

**c** Calculate the probability  $P(X < 3)$ .

**d** Using  $P(X = 4)$ , calculate the probability of success  $p$ .

**e** Calculate  $E(X)$ .

**f** Calculate the standard deviation of the distribution.

**24** The table represents a cumulative probability distribution of a binomial random variable:

| $x$           | 0       | 1       | 2       | 3       | 4       | 5 |
|---------------|---------|---------|---------|---------|---------|---|
| $P(X \leq x)$ | 0.01024 | 0.08704 | 0.31744 | 0.66304 | 0.92224 | 1 |

**a** State the number of trials for this distribution.

**b** Calculate  $P(X \leq 2)$ .

**c** Calculate  $P(X \geq 3)$ .

**d** Calculate  $P(X = 5)$ .

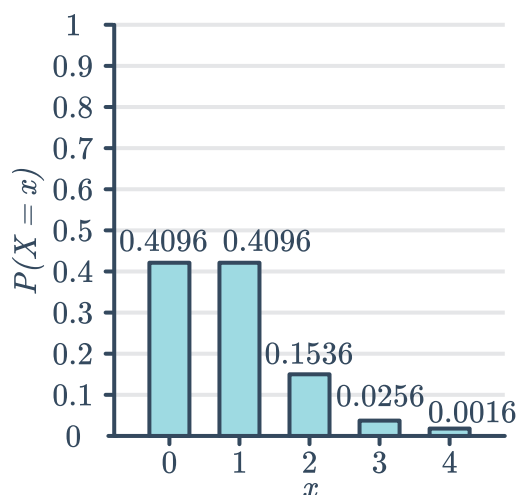
**e** Using  $P(X = 5)$ , calculate the probability of success  $p$ .

**f** Calculate the standard deviation of the distribution.

**25** The following graph represents a probability distribution of a binomial random variable:

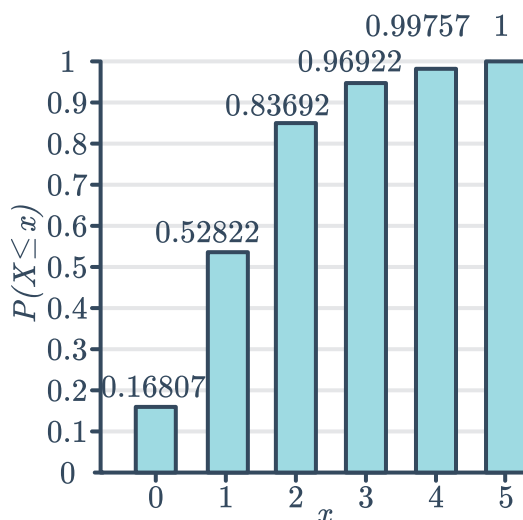


- a State the number of trials for this distribution.
- b Calculate  $P(X = 3)$ .
- c Calculate  $P(X < 3)$ .
- d Using  $P(X = 4)$ , calculate the probability of success  $p$ .
- e Calculate  $E(X)$ .
- f Calculate the standard deviation of the distribution.



**26** The following graph represents a cumulative probability distribution of a binomial random variable:

- a State the number of trials for this distribution.
- b Calculate  $P(X \leq 2)$ .
- c Calculate  $P(X \geq 3)$ .
- d Calculate  $P(X = 5)$ .
- e Using  $P(X = 5)$ , calculate the probability of success  $p$ .
- f Calculate the standard deviation of the distribution.



**27** Calculate the probability of the following, rounding your answers to two decimal places:

- a Getting exactly 18 successes in 30 independent trials if the probability of success in each trial is 0.7.
- b Getting at least 10 successes in 80 independent trials if the probability of success in each trial is  $p = 0.1$ .
- c Getting at most 22 successes in 25 independent trials if the probability of success in each trial is 0.85.
- d Getting more than 15 successes in 33 independent trials if the probability of success in each trial is 0.45.
- e Getting less than 12 successes in 50 independent trials if the probability of success in each trial is 0.28.

**28**  $P(X = 2) = 0.3456$  for a binomial random variable  $X$  with 4 trials.



- a** If success is more likely, find the probability of success  $p$ .
- b** Calculate  $E(X)$ .
- c** Calculate  $\text{Var}(X)$ .

**29**  $E(X) = 6$  for a binomial random variable  $X$  with 10 trials.

- a** Find the probability of success  $p$ .
- b** Calculate  $\text{Var}(X)$ .

**30** The expected value of a binomial distribution is 8 and the variance is 6.4.

- a** Find the number of trials,  $n$ .
- b** Find the probability of success,  $p$ .

**31**  $E(X) = \frac{15}{2}$  for a binomial random variable  $X$  with probability of success  $\frac{3}{10}$ .

- a** Find the number of trials  $n$ .
- b** Calculate the standard deviation of  $X$ .

**32**  $E(X) = 2$  and  $\text{Var}(X) = 1.6$  for a binomial random variable  $X$ .

- a** Find the probability of success  $p$ .
- b** Find the number of trials  $n$ .

**33**  $E(X) = 4.8$  and  $\sigma(X) = \sqrt{2.88}$  for a binomial random variable  $X$ .

- a** Find the probability of success  $p$ .
- b** Find the number of trials  $n$ .

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## Applications

**34** A certain disease has a survival rate of 64%. Of the next 110 people who contract the disease, how many would you expect to survive? Round your answer to the nearest whole number.

**35** The probability of a particular tennis player getting their first serve in is 0.72. If she serves 90 times in a match, how many times would she expect to get her first serve in? Round your answer rounded to the nearest whole number.

**36** A fair die is rolled 5 times and the number of fours noted.

- a** On each trial, what is the probability of success?
- b** Calculate the probability of rolling exactly 2 fours, correct to four decimal places.





- 37** David owns 3 yellow mugs and 8 blue mugs. Each morning, he chooses a mug at random for his cup of tea before work, 5 days a week. The mug is washed and returned before the next day.
- a** Each day, what is the probability of choosing a yellow mug?
  - b** What is the probability that he used a yellow mug exactly twice this past work week? Round your answer to four decimal places.
  - c** What is the probability that he never used a yellow mug this past work week? Round your answer to four decimal places.
- 38** At a local Italian restaurant, 16 people are in line to order. The probability that any one person will choose pizza is 23%.
- a** What is the probability that exactly 4 people choose pizza? Round your answer to four decimal places.
  - b** What is the probability that exactly half of the people choose pizza? Round your answer to four decimal places.
- 39** Consider the binomial  $\left(\frac{1}{5} + \frac{4}{5}\right)^5$ .
- a** Write the formula for the term of the expansion that contains  $\left(\frac{4}{5}\right)^3$ .
  - b** A box of chocolates is  $\frac{1}{5}$  dark chocolate and  $\frac{4}{5}$  milk chocolate. If 5 chocolates are selected at random with replacement, describe what the term from part (a) would tell us.
  - c** Calculate the probability of selecting exactly 1 dark chocolate.
  - d** Calculate the probability of selecting at most 1 dark chocolate.